# Lecture XIV - Recap and Discussion

Applied Optimization with Julia

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## Introduction

### Congratulations!

You've finished the course and learned how mathematical models can be used to solve real-world problems!

### Topics Covered

**Question: Can you recall the problem structure behind each topic?**

- Solar Panel Transport
- Beer Production
- Split Order Minimization
- Library Routing
- Police Service Districting
- Safety Planning for the Hajj Pilgrimage
- Arena Seating
- Passenger Flow Control

### Underlying Problem Structures

| Topic | Original Problem[1] |
|---|---|
| Solar Panel Transport | Classic Transport Problem |
| Beer Production | Capacitated Lot-Sizing Problem (CLSP) |
| Split Order Minimization | Quadratic Multiple Knapsack Problem (QMKP) |
| Library Routing | Capacitated Vehicle Routing Problem (CVRP) |
| Police Service Districting | p-Median Problem |
| Safety Planning for the Hajj Pilgrimage | Scheduling Problem |
| Arena Seating | 2D-Knapsack Problem |
| Passenger Flow Control | Dynamic Network Flow Problem |

[1] Original problem structure we **used/extended/exploited** to address the topic.

## Modeling Techniques

**Question: And do you remember the key modeling ideas?**

| Topic | Key Modeling Idea |
|---|---|
| Solar Panel Transport | Sets, parameters and continuous variables |
| Beer Production | Binary setup variables and Big-M constraints |
| Split Order Minimization | Quadratic objective with binary variables |
| Library Routing | Subtour elimination and heuristics |

## Modeling Techniques II

| Topic | Key Modeling Idea |
|---|---|
| Police Service Districting | Contiguity and compactness constraints |
| Safety Planning for the Hajj Pilgrimage | Time-indexed scheduling with penalties |
| Arena Seating | One binary variable per seating group |
| Passenger Flow Control | Queue dynamics over time periods |

## What have we learned?

- How to **identify and abstract** real-world problems
- How to **start programming** in Julia
- How to **model and solve** optimization problems
- How to **question** model assumptions

> **Tip**
> That's a lot and a great foundation for a seminar or a master thesis!

## Check Yourself

Before the exam, you should be able to answer the following:

- Can you **define sets, parameters, and variables** for a new problem?
- Can you **choose the right variable domain** (continuous, integer, binary)?
- Can you **formulate objective functions and constraints** in JuMP?
- Do you know **when a model needs Big-M constraints**?
- Can you **explain what subtours are** and how to prevent them?
- Do you know the **difference between LP, MIP, and NLP**?

> **Tip**
> If some points feel shaky, revisit the linked lectures and take a look at the cheatsheets on the course website.

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Any questions
regarding the
past lectures?

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# How to continue?

## How to continue after the lecture?

- The best way is to **keep programming and modeling**
- We offer **seminar places and master thesis supervision**
- Try to find a way to **apply programming in your work**
- There are **many interesting topics** to explore!

> **Tip**
> Getting your managers on board is the hardest part! But note that it is often worth it. The tools we have used are all free and open-source.

## Concrete Next Steps

- Ask questions on the Julia Discourse forum
- Explore the **JuMP community** at jump.dev
- Contribute to open-source, e.g. **"good first issues"** of JuMP
- Start a **small personal project** using the tools learned

> **Tip**
> Project ideas at the right scale: a schedule for your sports league, a weekly meal plan on a budget, or a shift roster for a student job.

## Start Pair Programming with AI

- First, try to **be confident with the basics** of a language
- Always try to **understand the code you use**
- Then, try an **AI-assisted IDE or coding agent**

> **Tip**
> The tools change fast; popular options are, for example, Cursor, GitHub Copilot, and Claude Code. They make work much easier compared to copying and pasting code between a chat and your editor.

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# Final Words

## That's it for the Lecture Series!

- I hope you **enjoyed** the lecture and found it **helpful**
- In the last tutorial, we will have a final **discussion session**
- There, you can earn the **last half-bonus point** for the exam
- I wish you **all the best for your studies and your career!**

> **Note**
> If you have any questions on optimization in the future, feel free to contact me!

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Questions?

## Thank You!

Thank you for participating in the course — good luck with the exam!

# Bibliography